

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 9: Supervised Learning

Day 62: Cross-validation for time series

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March 2026

1 Introduction

In machine learning, model validation is critical for evaluating how well a model generalizes to unseen data. A common method used in traditional machine learning is **k-fold cross-validation**. However, financial time series data introduces an important constraint:

Time has an order, and future information must never leak into the past.

Because of this temporal structure, standard cross-validation methods are not appropriate for financial data.

2 The Problem with Traditional Cross-Validation

In standard k-fold cross-validation, the dataset is randomly split into k parts.

The model is trained on $k - 1$ parts and tested on the remaining part.

Mathematically, if the dataset is

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

traditional cross-validation randomly partitions the data into folds.

This approach works for independent observations but fails for time series because:

- Financial data is **time dependent**
- Future observations must not influence past predictions
- Random shuffling causes **data leakage**

Data leakage occurs when the model accidentally learns information from the future.

3 Time Series Cross-Validation

To solve this issue, we use **time-aware validation techniques** that preserve chronological order.

One of the most common methods is:

Walk-Forward Validation

4 Walk-Forward Validation

In walk-forward validation, the training window expands over time, and the model is tested on future data.

Example structure:

Training Data	Test Data
1 – 100	101
1 – 101	102
1 – 102	103
1 – 103	104

Mathematically, if the dataset contains T observations:

Training set at step t :

$$D_{train} = \{(x_1, y_1), (x_2, y_2), \dots, (x_t, y_t)\}$$

Testing point:

$$D_{test} = (x_{t+1}, y_{t+1})$$

The prediction error is computed as

$$Error_{t+1} = (y_{t+1} - \hat{y}_{t+1})^2$$

Total model performance:

$$CV_{error} = \frac{1}{T-k} \sum_{t=k}^{T-1} (y_{t+1} - \hat{y}_{t+1})^2$$

This approach ensures that predictions are always made using only past information.

5 Rolling Window Validation

Another variation is the **rolling window approach**.

Instead of expanding the training set, we use a fixed-length window.

Example:

Training Window	Test Data
1 – 100	101
2 – 101	102
3 – 102	103
4 – 103	104

Mathematically:

$$D_{train} = \{(x_{t-w}, \dots, x_t)\}$$

where w is the window size.

This is useful in financial markets where old data may become less relevant.

6 Why Cross-Validation Matters in Finance

In financial modeling, proper validation is essential because:

- Markets evolve over time
- Models can easily overfit historical data
- Incorrect validation leads to unrealistic backtests

A model that performs well under improper validation may fail in real trading.
Proper cross-validation simulates how the model would behave in real-world deployment.

7 Application in Quantitative Finance

Time series cross-validation is used in many financial tasks:

- Return prediction models
- Volatility forecasting
- Algorithmic trading signals
- Risk forecasting

Example objective:

$$r_{t+1} = f(X_t) + \epsilon_t$$

Where:

- X_t = market features at time t
- r_{t+1} = next-period return
- $f(\cdot)$ = predictive model

The model must be trained only on past observations to ensure realistic evaluation.

8 Key Takeaways

- Standard cross-validation does not work for time series data
- Random shuffling causes data leakage
- Walk-forward validation preserves time order
- Rolling windows adapt to changing markets
- Proper validation is critical for reliable trading strategies